



Sequential Circuit

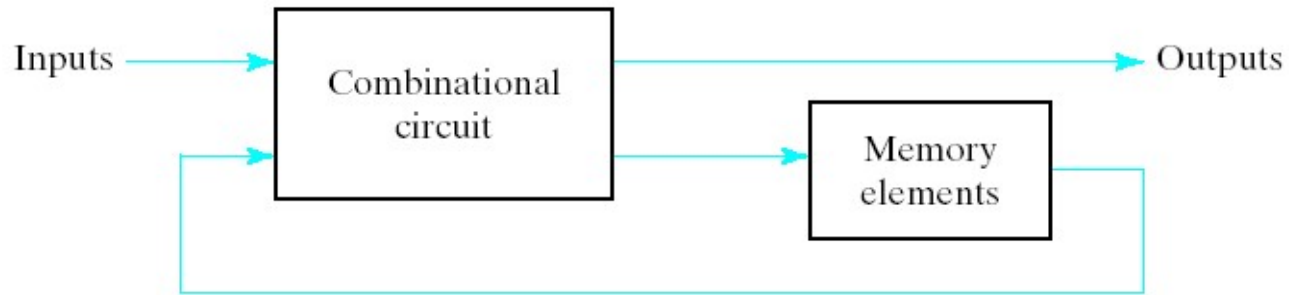


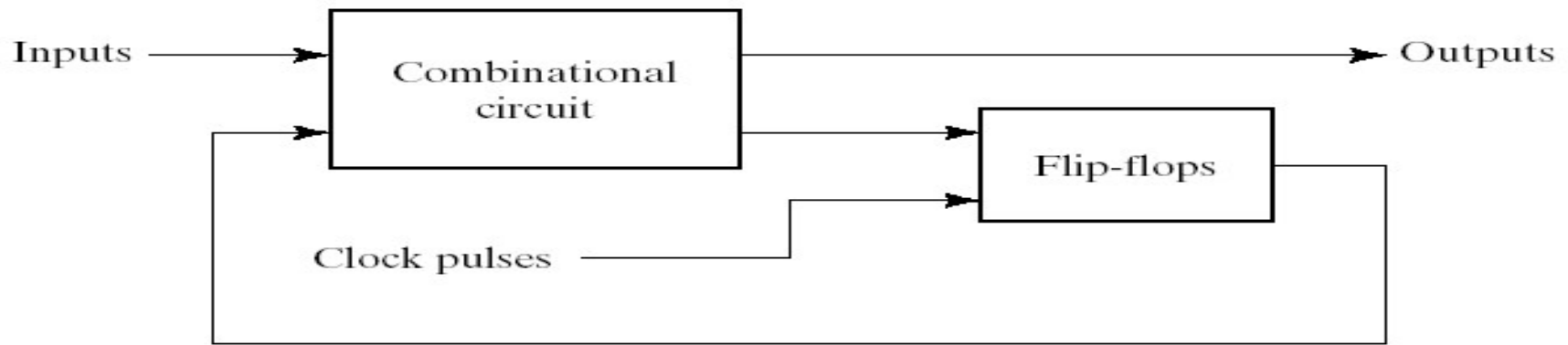
Fig. 5-1 Block Diagram of Sequential Circuit

- Outputs are functions of inputs and present states of storage elements
- Two types of sequential circuits
 - Synchronous (preferred !!)
 - Asynchronous



Synchronous Clocked Sequential Circuit

A **Synchronous sequential** circuit uses clock pulses for each memory element to hold / transmit the information.



(a) Block diagram



(b) Timing diagram of clock pulses

Fig. 5-2 Synchronous Clocked Sequential Circuit



Memory Elements

- Allow sequential logic design
- Latch — a level-sensitive memory element
 - SR latches
 - D latches
- Flip-Flop — an edge-triggered memory element
 - Master-slave flip-flop
 - Edge-triggered flip-flop
- RAM and ROM — a mass memory element



S-R Latch

The **SR latch** is a circuit with two cross-coupled NOR gates or two cross-coupled NAND gates. It has two inputs labeled S for set and R for reset.

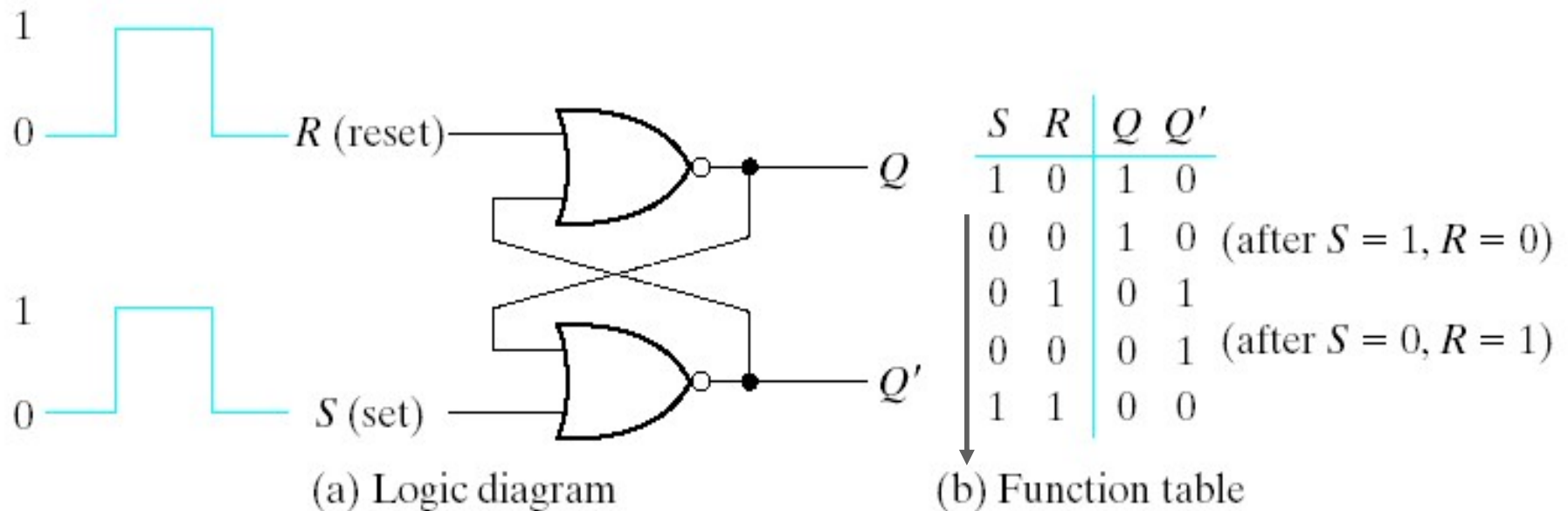


Fig. 5-3 SR Latch with NOR Gates



SR Latch with NAND Gates

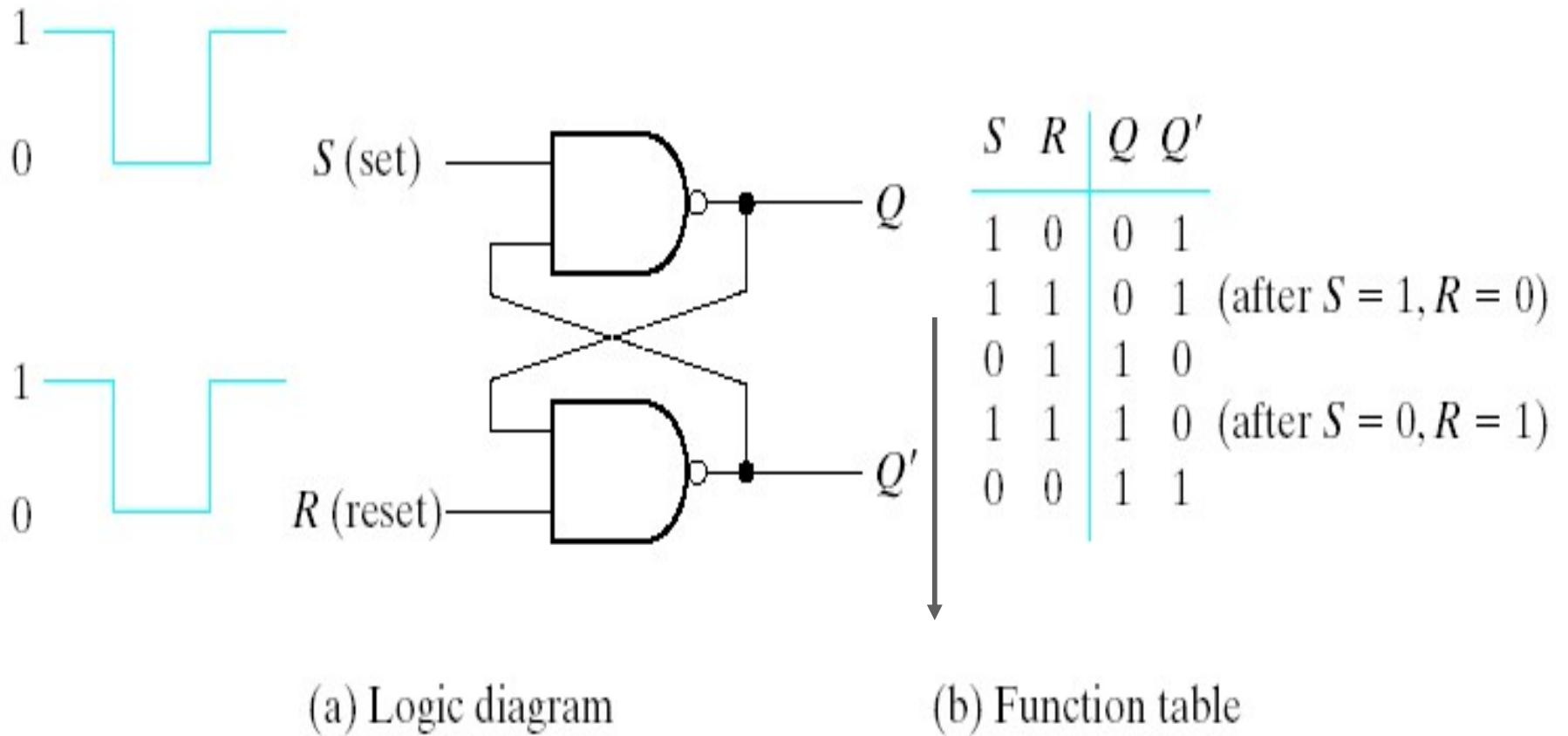
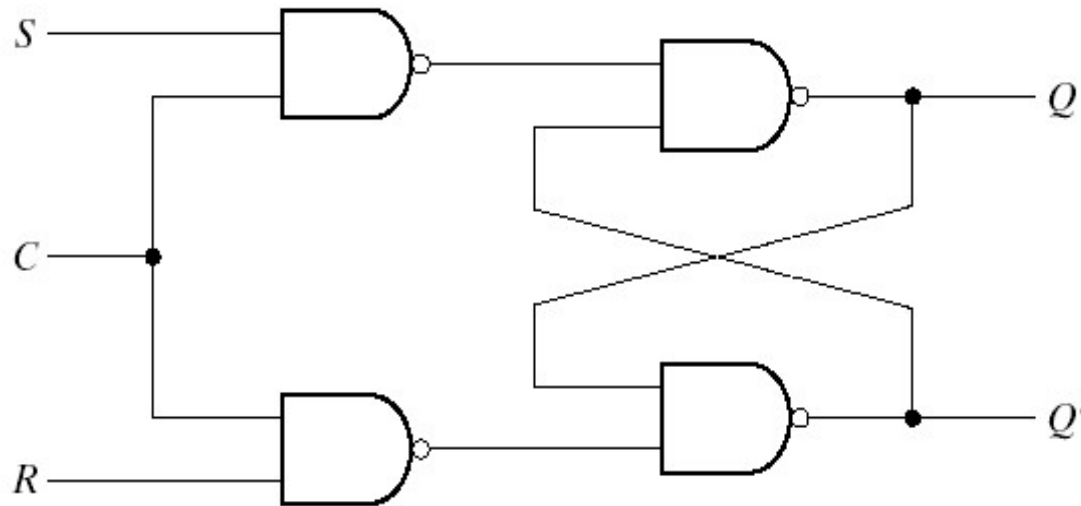


Fig. 5-4 SR Latch with NAND Gates



SR Latch with Control Input

The operation of the basic SR latch can be modified by providing an additional control input that determines when the state of the latch can be changed. In Fig. 5-5, it consists of the **basic SR latch** and **two additional NAND gates**.



(a) Logic diagram

C	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; Reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

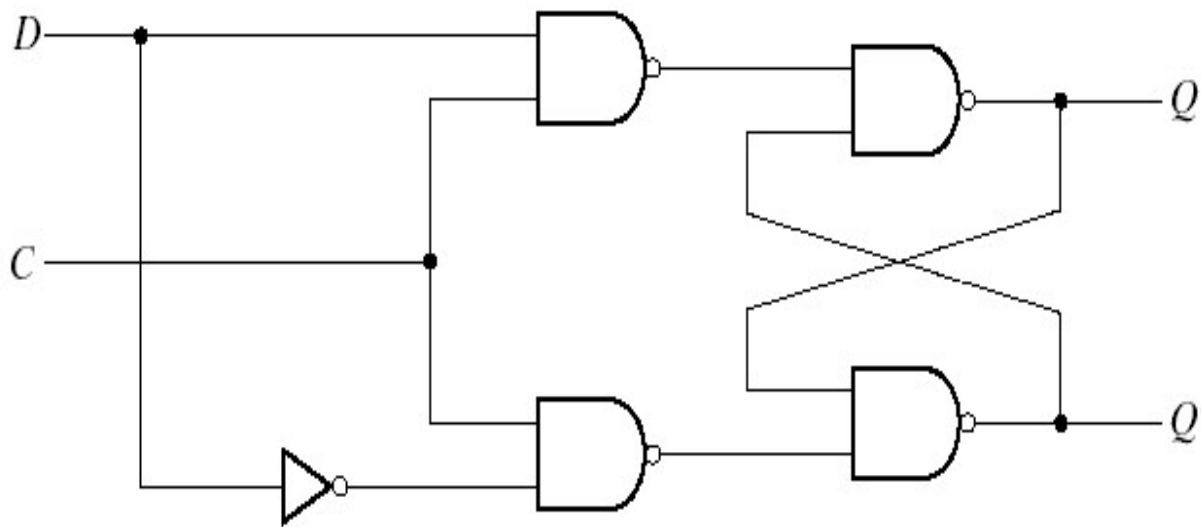
(b) Function table

Fig. 5-5 SR Latch with Control Input



D Latch

One way to eliminate the undesirable condition of the indeterminate state in SR latch is to ensure that **inputs S and R are never equal to 1 at the same time** in Fig 5-5. This is done in the D latch.



(a) Logic diagram

C	D	Next state of Q
0	X	No change
1	0	$Q = 0$; Reset state
1	1	$Q = 1$; Set state

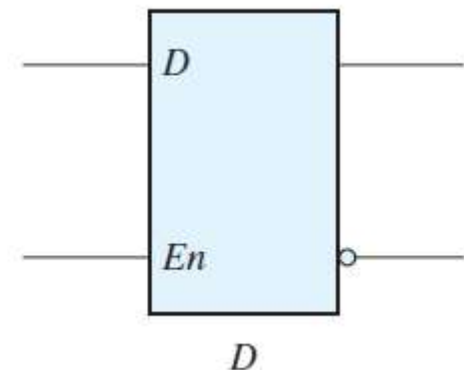
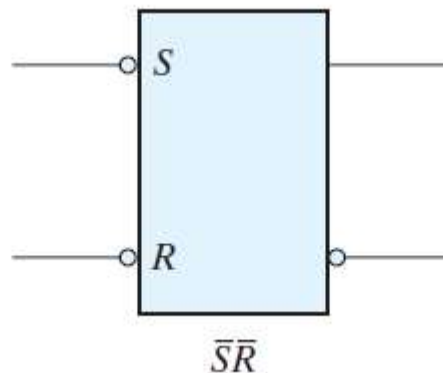
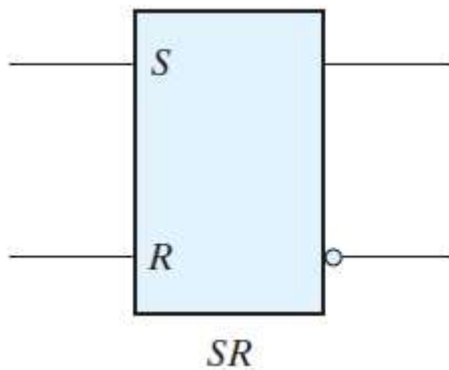
(b) Function table

Fig. 5-6 D Latch



Graphic Symbols for latches

A latch is designated by a rectangular block with inputs on the left and outputs on the right. One output designates the **normal output**, and the other designates the **complement output**.





Latch vs. Flip-Flop

Latch:

- Change stored value under specific status of the control signals
- Transparent for input signals when control signal is “on” in fig(a)

Flip-Flop

- Can only change stored value by a momentary switch in value of the control signals in fig (b)
- Cannot “see” the change of its output in the same clock pulse
- Encounter fewer problems than using latches



(a) Response to positive level

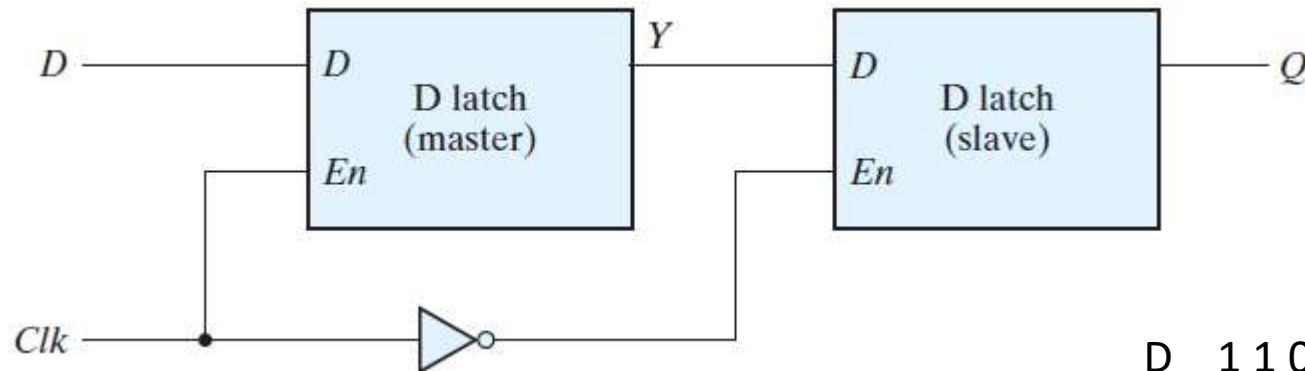


(b) Positive-edge response

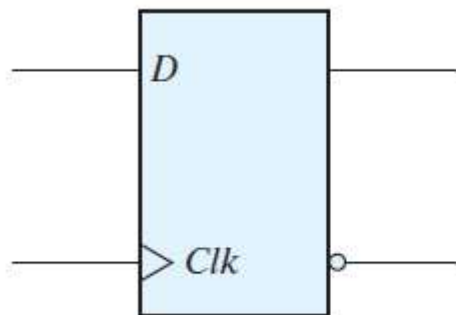
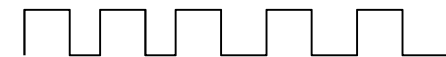


Master-Slave SR Flip-Flop

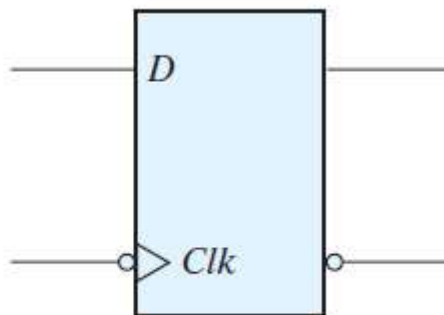
- The first latch is called the **master** and the second the **slave**. The circuit samples the D input and changes its output Q only at the negative-edge of the controlling clock.



D 1 1 0 0 1 1 ...
Y 1 1 0 0 1 1 ...
Q ? 1 1 0 0 1



(a) Positive-edge

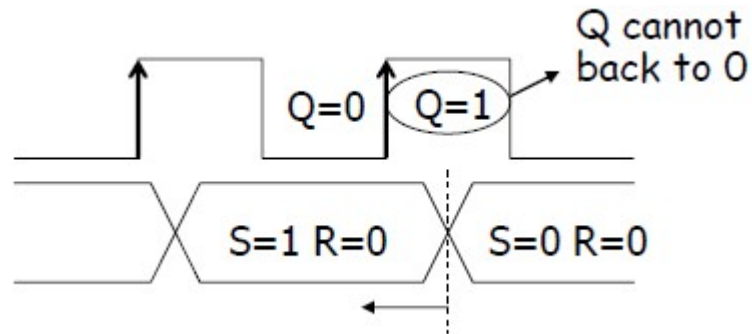


(a) Negative-edge



Problems of Master-Slave F/F

- Master-slave flip-flops are also referred as **pulse triggered** flip-flops
- May cause errors when the delay of combinational feedback path is too long



To solve:

- Ensure the delay of combinational block is short enough
- Use **edge-triggered** flip-flops instead



D- type Positive-Edge-Triggered F/F

- If only SR latches are available, three latches are required
- Two latches are used for locking the two inputs (CLK & D)
- The final latch provides the output of the flipflop

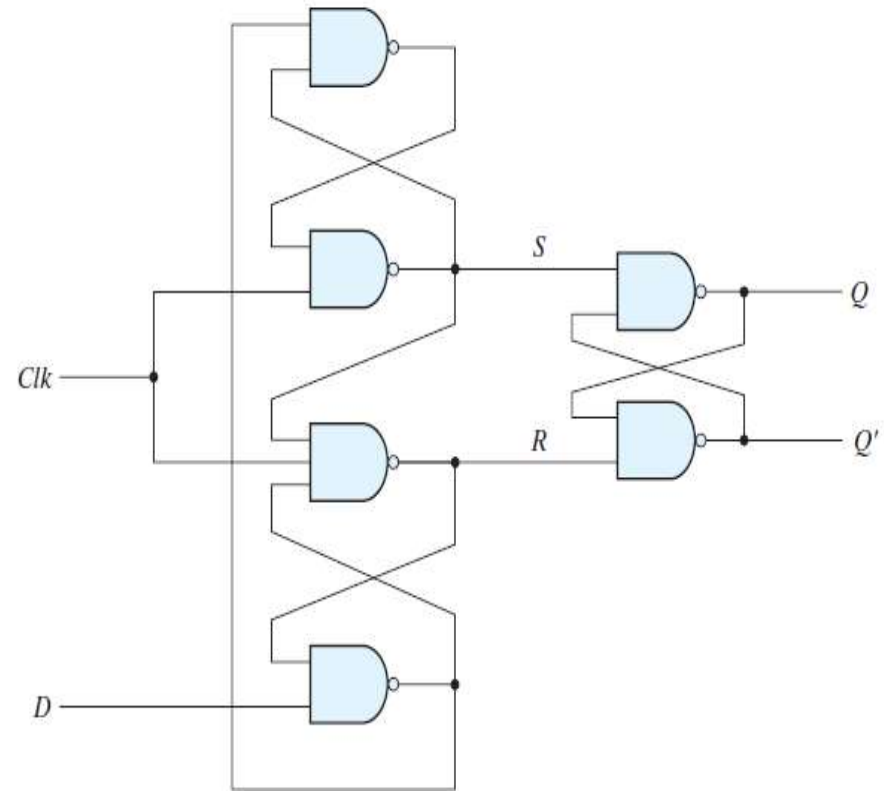


FIGURE 5.10

D-type positive-edge-triggered flip-flop



Flip-Flop Characteristic Tables

<i>JK</i> Flip-Flop			
<i>J</i>	<i>K</i>	$Q(t + 1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

<i>D</i> Flip-Flop		
<i>D</i>	$Q(t + 1)$	
0	0	Reset
1	1	Set

<i>T</i> Flip-Flop		
<i>T</i>	$Q(t + 1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement



Flip-Flop Tables

FLIP-FLOP NAME	FLIP-FLOP SYMBOL	CHARACTERISTIC TABLE	CHARACTERISTIC EQUATION	EXCITATION TABLE																																			
SR		<table> <tr> <th>S</th> <th>R</th> <th>$Q(next)$</th> </tr> <tr> <td>0</td> <td>0</td> <td>Q</td> </tr> <tr> <td>0</td> <td>1</td> <td>0</td> </tr> <tr> <td>1</td> <td>0</td> <td>1</td> </tr> <tr> <td>1</td> <td>1</td> <td>NA</td> </tr> </table>	S	R	$Q(next)$	0	0	Q	0	1	0	1	0	1	1	1	NA	$Q(next) = S + R'Q$ $SR = 0$	<table> <tr> <th>Q</th> <th>$Q(next)$</th> <th>S</th> <th>R</th> </tr> <tr> <td>0</td> <td>0</td> <td>0</td> <td>X</td> </tr> <tr> <td>0</td> <td>1</td> <td>1</td> <td>0</td> </tr> <tr> <td>1</td> <td>0</td> <td>0</td> <td>1</td> </tr> <tr> <td>1</td> <td>1</td> <td>X</td> <td>0</td> </tr> </table>	Q	$Q(next)$	S	R	0	0	0	X	0	1	1	0	1	0	0	1	1	1	X	0
S	R	$Q(next)$																																					
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T	$Q(next)$																																						
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